ALGEBRA 1 · UNIT 7: EXPONENTS AND SCIENTIFIC NOTATION

Scale of the Universe

Use scientific notation and exponent rules from viruses to stars.

Honors

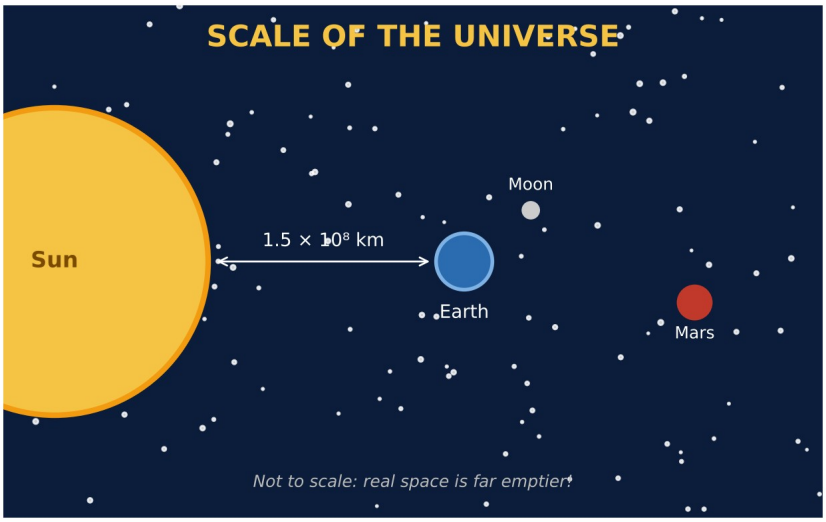
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

NASA has asked students to help explain space to younger kids. The problem: space is enormous, and the things inside our bodies are tiny. Numbers like 150,000,000 km and 0.00008 m are hard to picture.

You will use scientific notation and exponent rules to compare sizes, compute travel times, and then design a scale model of the solar system for your school hallway.



Learning Goals

- I can apply all exponent properties, including powers of products and negative exponents.
- I can place quantities on a logarithmic (powers-of-ten) scale.
- I can design a model using two scales and justify the choice.

Project Timeline

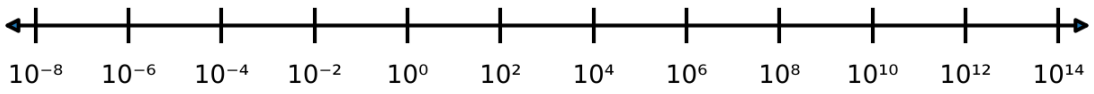
When	What you will do
Day 1	Powers of Ten
Day 2	Light & Travel
Day 3	Design a Two-Scale Exhibit
Final	Scale Model Plan + reflection

Part 1: Powers of Ten

Day 1 · Use the data cards. Convert every length to meters.

1. Place the flu virus, red blood cell, hair, Earth's diameter, Earth-Sun distance, and Proxima distance on the powers-of-ten line (estimate each power).

Size in meters (each tick is 100 times the one before)



2. Explain why equal spaces on this line represent multiplying by the same number, not adding.

3. Simplify $(2 \times 10^{3})^{3}$ and $(8 \times 10^{-5})^{2} \div (4 \times 10^{-7})$. Write answers in scientific notation.

4. A student wrote $(3 \times 10^{4})(5 \times 10^{-9}) = 15 \times 10^{-36}$. Identify both errors and correct the answer.

Part 2: Light & Travel

Day 2 · Light travels 3×10^5 km per second.

5. Find a light-year in km. Then express the Proxima Centauri distance in light-years.

Earth to Moon	Earth to Sun	Earth to Mars
3.84×10^5 km	1.5×10^8 km	5.6×10^7 km (closest)
Speed of light	**Earth's diameter**	**Mass of Earth**
3×10^5 km per second	1.27×10^4 km	5.97×10^{24} kg
Mass of Moon	**Human hair width**	**Red blood cell**
7.35×10^{22} kg	8×10^{-5} m	8×10^{-6} m
Flu virus	**Proxima Centauri**	**Seconds in a year**
1×10^{-7} m	4.0×10^{13} km away	$\approx 3.15 \times 10^7$ s

6. Find the volume of a cube-shaped box of side 8×10^{-6} m (about the size of a blood cell). How many such boxes fit in 1 cubic millimeter (10^{-9} m^3)?

7. Light from Proxima left the star how many seconds ago? Express it two ways: in seconds and in years.

8. If a probe could travel at 1% of the speed of light, how many years would it take to reach Proxima? Show how the exponent rules make this quick.

Part 3: Design a Two-Scale Exhibit

Day 3 · One scale shows distances in the hallway (50 m). A second scale shows sizes on a table.

9. Choose a distance scale so Mars (2.28×10^8 km from the Sun) is at 45 m or less. Find the model distances for Earth and Mars.

10. Choose a size scale so Earth's diameter is 10 cm. Find the model size of the Moon (diameter 3.47×10^3 km) and the Sun (diameter 1.39×10^6 km).

11. Explain why one scale cannot show both sizes and distances well. Use the ratio of the two scales.

Final Product: Scale Model Plan

Your final product must include:

- Completed powers-of-ten line
- Exponent-rule simplifications and error analysis
- Light-year and volume calculations
- Two scales with model sizes and distances
- Exhibit plan with explanation of the two scales

Reflection

R1. How do the exponent rules let you estimate huge calculations without a calculator?

R2. Why do scientists often use logarithmic scales for data that spans many powers of ten?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Model fits the chosen space and every scaled distance is calculated correctly. Goes beyond the minimum with a creative, realistic choice.	Model fits the chosen space and every scaled distance is calculated correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Scale model plan shows the scale, every converted distance in scientific notation, and a labeled diagram. Polished, organized, and visually engaging.	Scale model plan shows the scale, every converted distance in scientific notation, and a labeled diagram. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.